



EarStudio ES100 : Preamplifier

Application Note

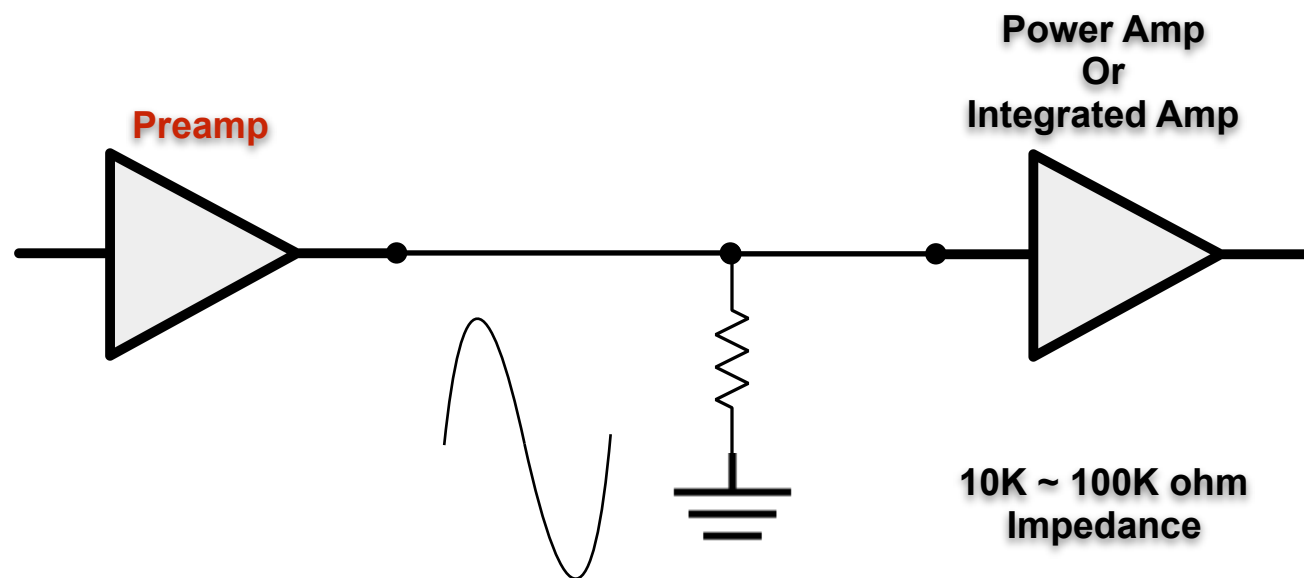
SEP 2018

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Preamplifier vs. Headphone Amplifier

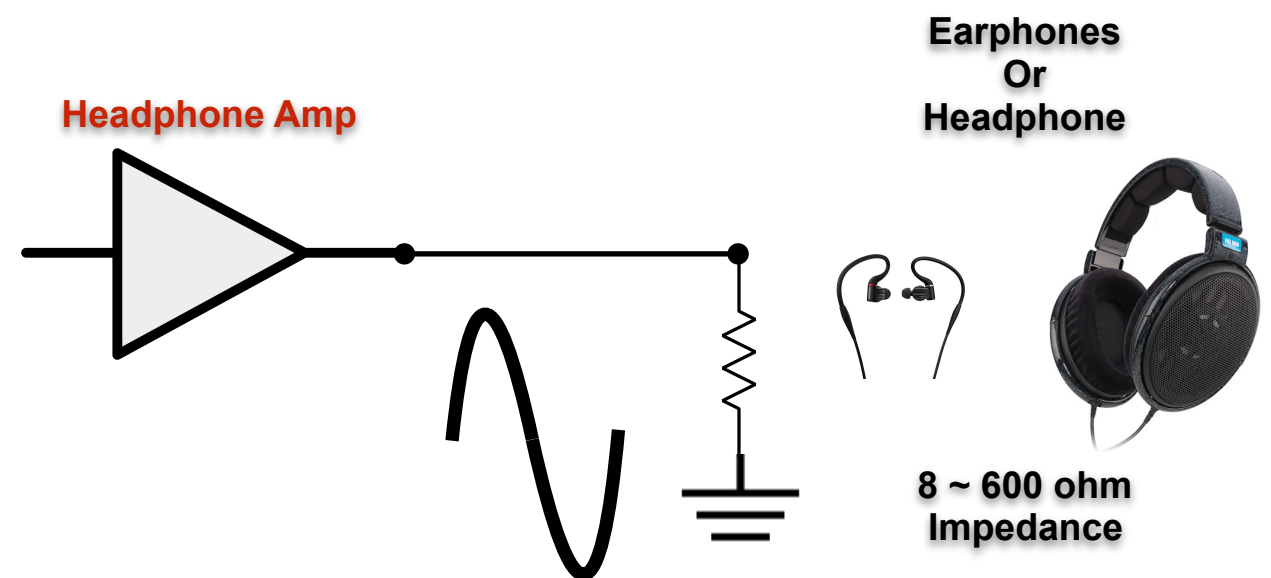
- Preamplifier

- drives 10K~100K ohm load(Power amplifier input impedance).
- Less power is required.
- Noise figure is always important since it can be amplified by the power amp.
- Need to check the output voltage level fit to the power amplifier input range.



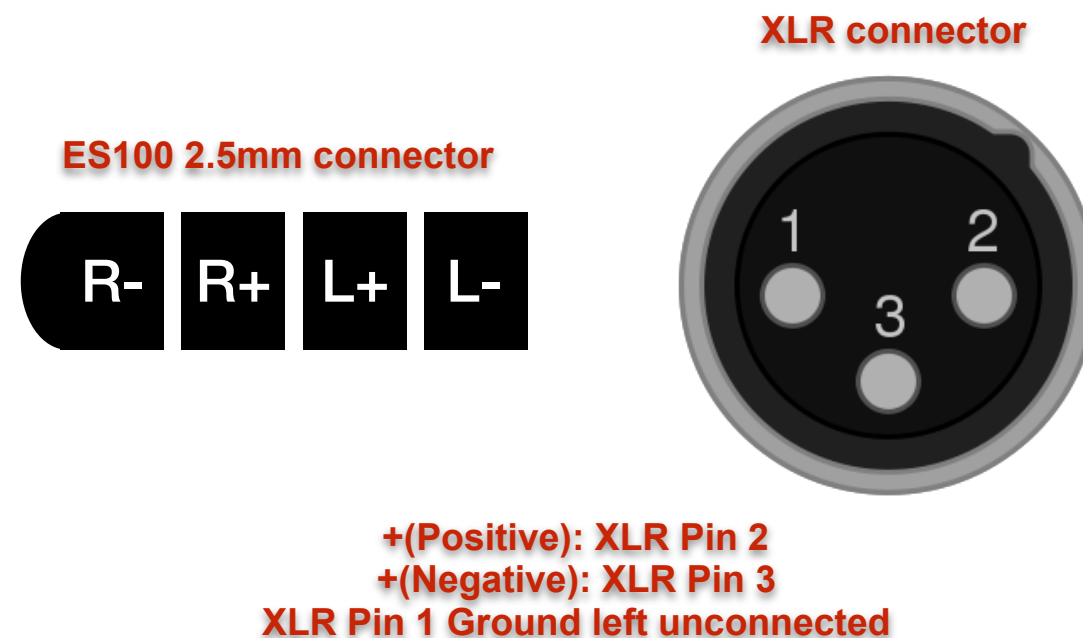
- Headphone Amplifier

- drives 8~600 ohm load.
- Noise figure is important for the high sensitivity unit.
- High Power is required for the low sensitivity unit.
- No need to check the voltage level / Just adjust the output level to the listening level.



ES100 Preamplifier Connection

- ES100 works as a preamplifier when connected to the other amplifier input.
- Connect Unbalanced(3.5mm) output to Unbalanced input / Balanced(2.5mm) output to Balanced input.
 - ***DO NOT connect the 3.5mm output to Balanced or the 2.5mm to Unbalanced, back and forth.**
 - The car aux-in mostly is 3.5mm female port and unbalanced.
- **The cable impedance doesn't matter in this case.**
 - Power amplifier or car aux-in, the input impedance is usually very high around 10K~100K.
 - The cable impedance won't affect any sound quality to that much of high input impedance.

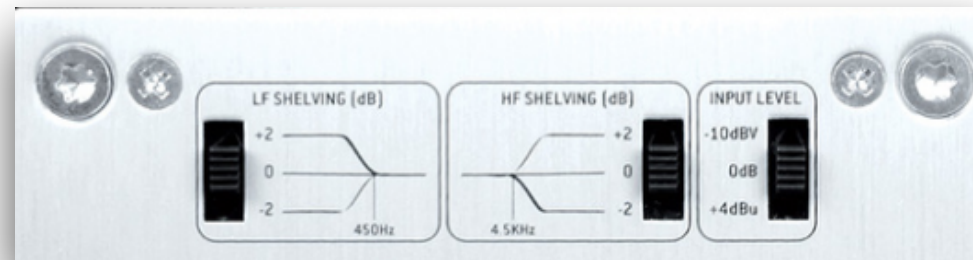


Line Level

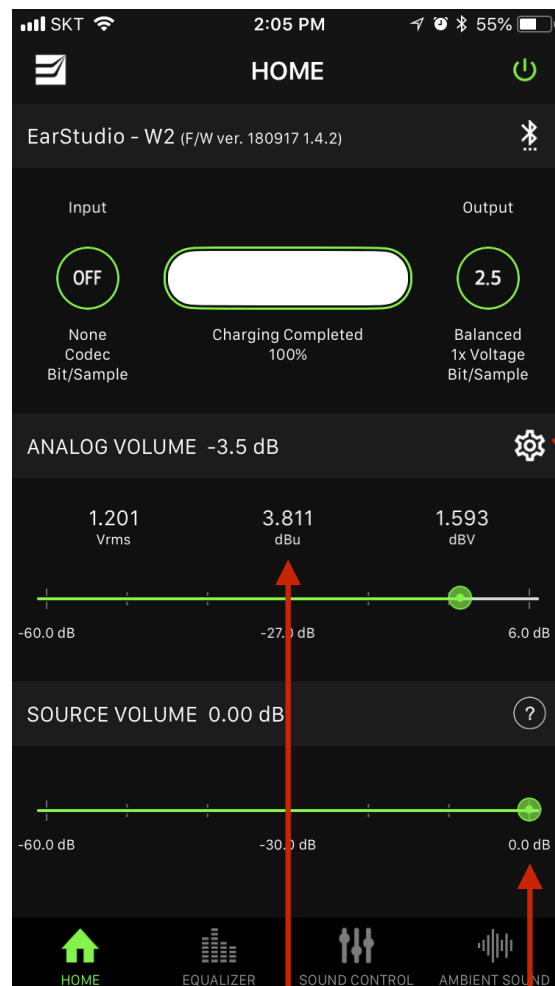
- Every amplifier input has its specific analog input range.
- It's essential to adjust ES100 output level fit to the input range of the connected amplifier.
- Refer the link below:
 - https://en.wikipedia.org/wiki/Line_level

Line levels and their approximate nominal voltage levels				
Use	Nominal level	Nominal level, V_{RMS}	Peak amplitude, V_{PK}	Peak-to-peak amplitude, V_{PP}
Professional audio	+4 dBu	1.228	1.736	3.472
Consumer audio	-10 dBV	0.316	0.447	0.894

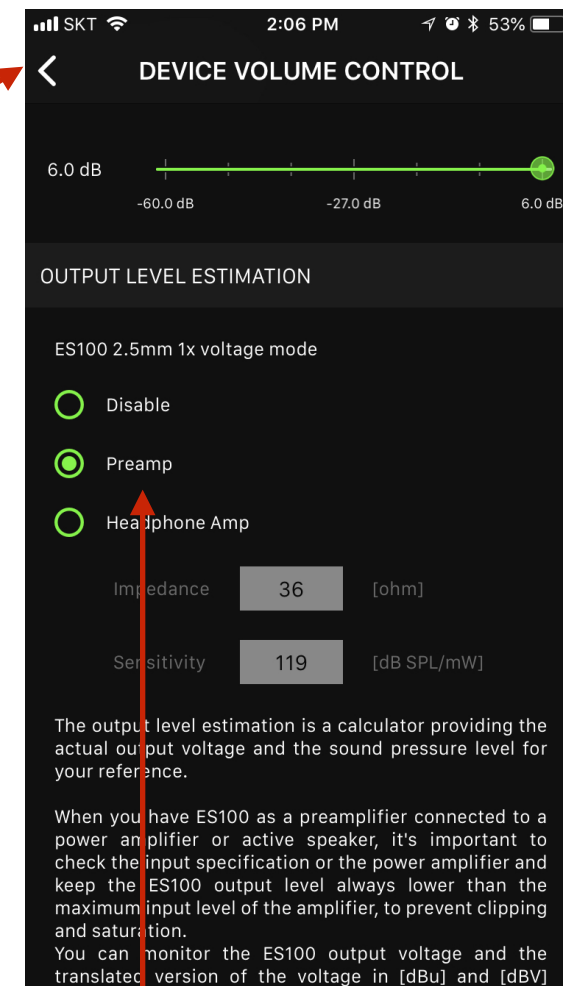
- Some active speakers or amplifiers have the input level selection option.



ES100 Output Level Estimation



**ES100 Output Level in Arms, dBu, and dBV.
All of them represent the same physical level.**



Select "Preamp" in Output Level Estimation.

Keep the source volume at the maximum for the optimal sound quality as well as the precise output level estimation.

Summary

- First off, check the aux-in or power amplifier input range specification.
 - Adjust ES100 output level fit to the input range, especially not to over and saturated.
 - Keep the source volume at the max. For the optimal sound quality and the output level estimation.
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- * ES100 Output Level Estimation option doesn't change the sound output. It's just a calculator to help you optimize your system setup.